

LOGICAL REASONING & PATTERN ANALYSIS

For Data Analysts and Computer Science Students

Introduction

Logical reasoning and pattern recognition assess your ability to identify relationships, trends, and hidden rules within data. These skills are frequently tested in:

- Data Analyst Aptitude Tests
- Graduate Trainee Assessments
- Technical Interviews
- Business Intelligence Roles
- Programming and Algorithm Challenges

The goal is to determine the underlying rule governing a sequence, set of numbers, or diagram and use it to predict the next value.

1. ARITHMETIC SEQUENCES

An arithmetic sequence changes by adding or subtracting the same value repeatedly.

Formula

Next Term = Previous Term + Common Difference

Example 1

Sequence:

2, 5, 8, 11, 14, ?

Find the difference:

$$5 - 2 = 3$$

$$8 - 5 = 3$$

$$11 - 8 = 3$$

$$14 - 11 = 3$$

Pattern:

+3

Answer:

$$14 + 3 = 17$$

Example 2

100, 95, 90, 85, 80, ?

Pattern:

-5

Answer:

75

Practical Data Analyst Example

Monthly customer growth:

Month	Customers
--------------	------------------

January	1000
---------	------

February	1200
----------	------

March	1400
-------	------

April	1600
-------	------

Pattern:

+200 customers monthly

Prediction:

May = 1800 customers

2. GEOMETRIC SEQUENCES

A geometric sequence changes by multiplying or dividing by a constant value.

Formula

Next Term = Previous Term $\times$ Common Ratio

Example 1

3, 6, 12, 24, 48, ?

Pattern:

$\times 2$

Answer:

96

Example 2

243, 81, 27, 9, 3, ?

Pattern:

$\div 3$

Answer:

1

Practical Data Analyst Example

Weekly website traffic doubles:

Week Visits

1 1000

2 2000

3 4000

4 8000

Prediction:

Week 5 = 16,000 visits

3. FIBONACCI PATTERNS

Each term equals the sum of the previous two terms.

Example

0, 1, 1, 2, 3, 5, 8, 13, ?

Calculation:

$$8 + 13 = 21$$

Answer:

21

Computer Science Applications

Fibonacci sequences appear in:

- Dynamic Programming
- Recursion
- Tree Algorithms
- Complexity Analysis

Common coding interview topic.

4. DIFFERENCE ANALYSIS

When a sequence appears irregular, calculate differences between consecutive numbers.

Example

3, 7, 13, 21, 31, ?

Differences:

4, 6, 8, 10

Observe:

Differences increase by 2

Next Difference:

12

Answer:

$$31 + 12 = 43$$

Example

2, 5, 10, 17, 26, ?

Differences:

3, 5, 7, 9

Next Difference:

11

Answer:

$$26 + 11 = 37$$

5. SQUARE NUMBER PATTERNS

Recognize perfect squares.

Sequence:

1, 4, 9, 16, 25, ?

These represent:

$$1^2, 2^2, 3^2, 4^2, 5^2$$

Answer:

$$6^2 = 36$$

Common Squares

Number Square

1	1
2	4
3	9
4	16
5	25
6	36
7	49
8	64
9	81
10	100

6. CUBE NUMBER PATTERNS

Recognize perfect cubes.

Sequence:

1, 8, 27, 64, 125, ?

These represent:

$1^3, 2^3, 3^3, 4^3, 5^3$

Answer:

$6^3 = 216$

Common Cubes

Number Cube

1	1
2	8
3	27
4	64
5	125
6	216

7. ALTERNATING PATTERNS

Some sequences contain two interwoven patterns.

Example

1, 4, 2, 8, 3, 12, 4, ?

Split into two groups:

First:

1, 2, 3, 4

Second:

4, 8, 12, ?

Pattern:

+4

Answer:

16

Example

2, 6, 3, 9, 4, 12, 5, ?

Pattern:

Two sequences:

2, 3, 4, 5

6, 9, 12, ?

Answer:

15

8. PERCENTAGE GROWTH PATTERNS

Very common in business analytics.

Example

Revenue:

1000, 1100, 1210, 1331, ?

Observe:

Each value increases by 10%.

$$1000 \times 1.1 = 1100$$

$$1100 \times 1.1 = 1210$$

$$1210 \times 1.1 = 1331$$

Answer:

$$1331 \times 1.1$$

$$= 1464.1$$

Real-World Application

Used for:

- Revenue Forecasting
- Customer Growth Prediction

- Market Expansion Analysis
-

9. MATRIX REASONING

Frequently appears in aptitude tests.

Example

2 4

6 ?

Observe:

$$2 \times 2 = 4$$

$$6 \times 2 = 12$$

Answer:

12

Example

3 6

5 10

8 ?

Pattern:

Multiply by 2

Answer:

16

10. DEDUCTIVE LOGICAL REASONING

Tests your ability to draw conclusions from statements.

Example

All Data Analysts use data.

Sarah is a Data Analyst.

Conclusion:

Sarah uses data.

Example

All servers are computers.

Some computers are laptops.

Can we conclude all servers are laptops?

Answer:

No.

There is insufficient information.

11. ANALYTICAL REASONING FOR DATA ANALYSTS

Real-world reasoning involves identifying trends and making predictions.

Example

Daily Active Users:

1000

1200

1400

1600

1800

Pattern:

+200

Prediction:

2000

Example

Monthly Revenue:

₦500,000

₦600,000

₦720,000

₦864,000

Pattern:

20% growth

Prediction:

₦1,036,800

QUICK APPROACH TO ANY PATTERN QUESTION

When solving a sequence:

Step 1: Check Addition/Subtraction

Step 2: Check Multiplication/Division

Step 3: Calculate Differences

Step 4: Look for Squares

Step 5: Look for Cubes

Step 6: Check Percentage Growth

Step 7: Look for Alternating Patterns

Step 8: Check Fibonacci Relationships

Step 9: Identify Real-World Trends

INTERVIEW TIPS

Common Data Analyst aptitude tests focus on:

Number Series

Pattern Recognition

Trend Analysis

Deductive Reasoning

Matrix Logic

Percentage Growth Patterns

Forecasting Questions

Data Interpretation-Based Logic